CHE-311

2-16-26

Problem Set 2

2.) Momentum has dimensions: $[p]=[mv]=MLT^{-1}$

Momentum per unit area per unit time: $\left[\frac{p}{At}\right]=\frac{MLT^{-1}}{L^2T}=ML^{-1}T^{-2}$

Force dimensions: $[F]=MLT^{-2}$

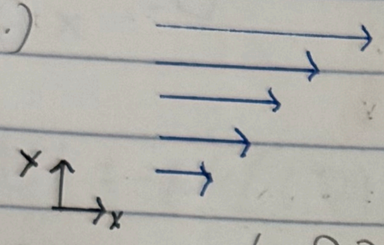
$\left[\frac{F}{A}\right]=\frac{MLT^{-2}}{L^2}=ML^{-1}T^{-2}$

$\frac{\text{Momentum}}{\text{Area}\cdot\text{time}}$ or $\frac{\text{Force}}{\text{Area}}$

3.) Molecular: Momentum that spreads due to viscosity

Convective: Momentum that is carried by a moving fluid

B.1.) a.)



Normal stress components.

$\tau_{xx}=-\mu\left(2\frac{\partial v_x}{\partial x}\right)=-\mu\left(2\frac{\partial}{\partial x}(by)\right)=0$

$\tau_{yy}=-\mu\left(2\frac{\partial v_y}{\partial y}\right)=-\mu\left(2\frac{\partial}{\partial y}(0)\right)=0$

$\tau_{zz}=-\mu\left(2\frac{\partial v_z}{\partial z}\right)=-\mu\left(2\frac{\partial}{\partial z}(0)\right)=0$

Shear stress component $\tau_{xy}=-\mu\left(\frac{\partial v_y}{\partial x}+\frac{\partial v_x}{\partial y}\right)=-\mu\left(\frac{\partial}{\partial x}(0)+\frac{\partial}{\partial y}(by)\right)=-\mu b$

$\tau_{yz}=-\mu\left(\frac{\partial v_z}{\partial y}+\frac{\partial v_y}{\partial z}\right)=-\mu\left(\frac{\partial}{\partial y}(0)+\frac{\partial}{\partial z}(0)\right)=0$

$\tau_{zx}=-\mu\left(\frac{\partial v_x}{\partial z}+\frac{\partial v_z}{\partial x}\right)=-\mu\left(\frac{\partial}{\partial z}(by)+\frac{\partial}{\partial x}(0)\right)=0$

Fluid velocity profile

X-direction

$\rho v_x v_x=\rho b^2 y^2$

$\rho v_y v_x=0$

$\rho v_z v_x=0$

$\rho v_x v_z=0$

X-direction $\rho v_y v_z=0$

$\rho v_z v_z=0$

Y-direction

$\rho v_x v_y=0$

$\rho v_y v_y=0$

$\rho v_z v_y=0$

b.)

Normal stress component $\tau_{xx} = -\mu\left(2\frac{\partial v_x}{\partial x}\right) = -\mu\left(2\frac{\partial}{\partial x}(bx)\right) = 0$

$\tau_{yy} = -\mu\left(2\frac{\partial v_y}{\partial y}\right) = -\mu\left(2\frac{\partial}{\partial y}(bx)\right)$

$\tau_{zz} = -\mu\left(2\frac{\partial v_z}{\partial z}\right) = -\mu\left(2\frac{\partial}{\partial z}(0)\right) = 0$

Shear Stress Component $\tau_{xy} = -\mu\left(\frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y}\right) = -\mu\left(\frac{\partial}{\partial x}(bx) + \frac{\partial}{\partial y}(by)\right) = -2\mu b$

$\tau_{yz} = -\mu\left(\frac{\partial v_z}{\partial y} + \frac{\partial v_y}{\partial z}\right) = -\mu\left(\frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(bx)\right) = 0$

$\tau_{zx} = -\mu\left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x}\right) = -\mu\left(\frac{\partial}{\partial z}(by) + \frac{\partial}{\partial x}(0)\right) = 0$

Fluid velocity profiles

X-direction $\rho v_x v_x = \rho b^2 y^2$ $\rho v_x v_y = \rho d^2 yx$

$\rho v_y v_x = \rho b^2 yx$

Y-direction $\rho v_y v_y = \rho d^2 x^2$

$\rho v_z v_x = 0$ $\rho v_z v_y = 0$

$\rho v_x v_z = 0$

Z-direction $\rho v_y v_z = 0$

$\rho v_z v_z = 0$

c.)

Normal stress component $\tau_{xx} = -\mu\left(2\frac{\partial v_x}{\partial x}\right) = -\mu\left(2\frac{\partial}{\partial x}(by)\right) = 0$

$\tau_{yy} = -\mu\left(2\frac{\partial v_y}{\partial y}\right) = -\mu\left(2\frac{\partial}{\partial y}(bx)\right) = 0$

$\tau_{zz} = -\mu\left(2\frac{\partial v_z}{\partial z}\right) = -\mu\left(2\frac{\partial}{\partial z}(0)\right) = 0$

Shear stress component $\tau_{xy} = -\mu\left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x}\right) = \mu\left(\frac{\partial}{\partial x}(bx) + \frac{\partial}{\partial y}(-by)\right) = 0$

component

$\tau_{yz} = -\mu\left(\frac{\partial v_z}{\partial y} + \frac{\partial v_y}{\partial z}\right) = -\mu\left(\frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(bx)\right) = 0$

$\tau_{xz} = -\mu\left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x}\right) = -\mu\left(\frac{\partial}{\partial z}(-by) + \frac{\partial}{\partial x}(0)\right) = 0$

fluid velocity profiles

x-direction $\rho v_x v_x = \rho b^2 y^2$

$\rho v_y v_x = -\rho b^2 yx$

$\rho v_z v_x = 0$

$\rho v_x v_z = 0$

y-direction $\rho v_x v_y = -\rho b^2 yx$

$\rho v_y v_y = \rho b^2 x^2$

$\rho v_z v_y = 0$

z-direction $\rho v_x v_z = 0$

$\rho v_y v_z = 0$

$\rho v_z v_z = 0$

d.) Normal stress component $\tau_{xx} = -\mu\left(2\frac{\partial v_x}{\partial x}\right) = -\mu\left(2\frac{\partial}{\partial x}\left(-\frac{1}{2}bx\right)\right) = \mu b$

$\tau_{yy} = -\mu\left(2\frac{\partial v_y}{\partial y}\right) = -\mu\left(2\frac{\partial}{\partial y}\left(-\frac{1}{2}by\right)\right) = \mu b$

$\tau_{zz} = -\mu\left(2\frac{\partial v_z}{\partial z}\right) = -\mu\left(2\frac{\partial}{\partial z}(bz)\right) = -\mu b$

Shear stress component $\tau_{xy} = -\mu\left(\frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y}\right) = -\mu\left(\frac{\partial}{\partial x}\left(-\frac{1}{2}by\right) + \frac{\partial}{\partial y}\left(-\frac{1}{2}bx\right)\right) = 0$

$\tau_{yz} = -\mu\left(\frac{\partial v_z}{\partial y} + \frac{\partial v_y}{\partial z}\right) = -\mu\left(\frac{\partial}{\partial y}(bz) + \frac{\partial}{\partial z}\left(-\frac{1}{2}by\right)\right) = 0$

$\tau_{zx} = -\mu\left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x}\right) = -\mu\left(\frac{\partial}{\partial z}\left(-\frac{1}{2}bx\right) + \frac{\partial}{\partial x}(bz)\right) = 0$

Velocity profile

x-direction $\rho v_x v_x = \rho \frac{1}{4} b^2 x^2$

$\rho v_y v_x = \rho \frac{1}{4} b^2 xy$

$\rho v_z v_x = -\rho \frac{1}{2} b^2 xz$

$\rho v_x v_z = -\rho \frac{1}{2} b^2 xz$

z-direction $\rho v_y v_y = -\rho \frac{1}{2} b^2 yz$

$\rho v_z v_z = \rho b^2 z^2$

y-direction $\rho v_x v_y = \rho \frac{1}{4} b^2 xy$

$\rho v_y v_y = \rho \frac{1}{4} b^2 y^2$

$\rho v_z v_z = -\rho \frac{1}{2} b^2 yz$